THE PULLBACK ALONG A FLAT FINITELY PRESENTED MORPHISM WITH GEOMETRICALLY REDUCED FIBRES COMMUTES WITH PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a good object with the required factorisation, the displayed five-term chain with the reason for each equality, and the covering argument giving the statement for all objects.

Editorial note. The label lemma-fppf-reduced-fibres-pullback-products names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Sections upstairs for a morphism with geometrically connected fibres. Same chapter, the lemma computing sections of a pullback on an object U whose structure morphism to the base is flat with geometrically connected fibres. Supplies the first and third equalities of the displayed chain.
- Every etale object admits a covering by such good objects. More on Morphisms, the lemma producing an etale covering of any object by pieces that factor through an etale morphism with geometrically connected flat finitely presented fibres. Supplies the reduction from a single object to a covering, which gives the statement for all objects.
- Locality of the pullback along a morphism of sites. Sites, the lemma identifying the restriction of f^{-1}G to U_etale with the pullback along U -> V of the restriction of G. Supplies the reduction of the first and third equalities to the localised situation.

Source reference. Original reference 0A3H: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5030–5030. Prerequisite pointer invoked at proof line(s) 16771; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0EY5: sources/stacks-chow/tags.txt, source lines 18680–18680. Prerequisite pointer invoked at proof line(s) 16778; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EF: sources/stacks-chow/tags.txt, source lines 4454–4454. Prerequisite pointer invoked at proof line(s) 16776; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation with geometrically reduced fibres. Then $f^{-1} : Sh(S_{étale}) \to Sh(X_{étale})$ commutes with products.*

Proof. Let I be a set and let $\mathcal{G}_i$ be a sheaf on $S_{étale}$ for $i \in I$. Let $U \to X$ be an étale morphism such that $U \to S$ factors as $U \to V \to S$ with $V \to S$ étale and $U \to V$ flat of finite presentation with geometrically connected fibres. Then we have

$$\begin{aligned} f^{-1}(\prod \mathcal{G}_i)(U) &= (\prod \mathcal{G}_i)(V) \\ &= \prod \mathcal{G}_i(V) \\ &= \prod f^{-1}\mathcal{G}_i(U) \\ &= (\prod f^{-1}\mathcal{G}_i)(U) \end{aligned}$$

where we have used Lemma 0A3H in the first and third equality (we are also using that the restriction of $f^{-1}\mathcal{G}$ to $U_{étale}$ is the pullback via $U \to V$ of the restriction of $\mathcal{G}$ to $V_{étale}$, see Sites, Lemma 03EF). By More on Morphisms, Lemma 0EY5 every object U of $X_{étale}$ has an étale covering $\{U_i \to U\}$ such that the discussion in the previous paragraph applies to U_i. The lemma follows. □